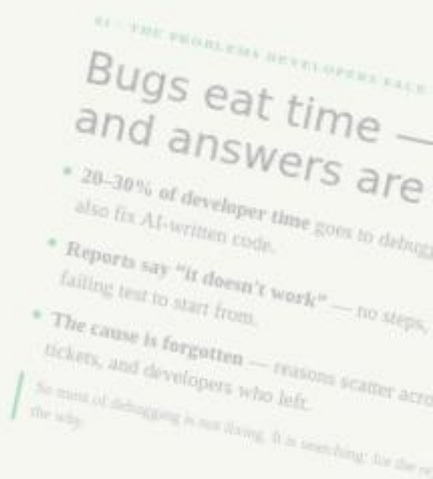
AGENTX HACKATHON 2026 · PS-05 BUG-FIX AGENT

CodeStitch

The multi-bug fixer that understands WHY before it stitches.

Team FORTEC Ronit Shrimankar (lead) · Anshul Kamthe

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DESTITCH — BY FORTEC

01 · THE PROBLEMS DEVELOPERS FACE

Bugs eat time — and answers are missing

- **20–30% of developer time** goes to debugging — and 95% now also fix AI-written code.
- **Reports say “it doesn't work”** — no steps, no environment, no failing test to start from.
- **The cause is forgotten** — reasons scatter across old commits, tickets, and developers who left.

> So most of debugging is not fixing. It is searching: for the repro, for the cause, for the why.

02 · WHAT TODAY'S FIXERS GET WRONG

Confident fixes that are wrong

- **Symptom fixes** — the error is hidden with a null check; the log goes quiet, the bug stays.
- **Fake-green fixes** — the patch passes old tests but misses what the code was supposed to do.
- **Regressions** — a fix in one place silently breaks features somewhere else.
- **Slow, costly, untrusted** — endless loops, burnt tokens, unsafe code, and a human must re-check all of it.

8 agents, one pipeline, one coordinator

1. **Triager** — real and reproducible? Rejects vague ones early.
2. **Localizer** — graph-guided to the exact method.
3. **Reproducer** — sandboxed failing test first. No repro, no fix.
4. **Parallel Fixers** — several candidates, isolated sandboxes.
5. **Adversarial Tester** — NEW intent tests, catch overfitting.
6. **Regression Gate** — targeted tests in loop, full suite at end.
7. **Reviewer + Human Gate** — human approves before merge.

> Coordinator on top: orders multi-bug repairs by dependency.

How CodeStitch solves each one

Vague reports

Triager asks clarifying questions, rejects early.

Unknown cause

History graph finds origin commit + discussion.

Symptom fixes

Repro-first: prove it, then fix the cause.

Fake-green fixes

New intent tests + independent verification.

Regressions

Targeted tests in loop, full suite at merge.

Slow & untrusted

Parallel fixers, tiered models, sandbox, human gate.

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CODESTITCH — BY FORTEC

Stitch the fix, not the symptom.

Build round: **Nov 11–12** · Proof: resolve rate · time-to-PR · regression rate · cost per fix.

Powered by the WhyCode history graph — our own engine, already proven on a large real-world repo.

